# Chapter 2: Generalized Linear Models

*One concept, one exercise per slide — practice as you go*

# Concept: The Three Components of a GLM

- Random component: the response distribution (e.g. Bernoulli, Poisson, Gamma)
- Systematic component: the linear predictor $\eta = X\beta$ (same as OLS)
- Link component: a function g connecting the mean to the linear predictor, $g(\mu) = \eta$
- OLS is the special case: identity link, Gaussian random component

✏️ **Exercise: A model predicts number of customer complaints per week (a count, always $\geq 0$). Explain why an ordinary linear regression's identity link is a poor systematic choice here, in terms of what values $X\beta$ can take versus what values the outcome can take.**

# Concept: The Exponential Family

- A distribution family with the form $f(y;\theta,\varphi) = \exp[(y\theta - b(\theta))/a(\varphi) + c(y,\varphi)]$
- $\theta$ is the natural parameter; $b(\theta)$ determines the mean and variance
- Normal, Bernoulli, Poisson, and Gamma are all members of this one family
- This unification is why one algorithm (IRLS) fits every GLM in this chapter

✏️ **Exercise: Why does writing every response distribution in the same exponential-family form make it possible to build one shared fitting algorithm instead of a different one for Poisson, logistic, and Gamma regression?**

# Concept: Link Functions and Canonical Links

- A link function $g$ maps the mean $\mu$ (restricted range) to $\eta$ (any real number)
- Canonical link: the link equals the natural parameter $\theta$, which simplifies the score equations
- Gaussian → identity; Poisson → log; Binomial → logit
- Non-canonical links are valid too (e.g. probit for binomial) — just less algebraically tidy

✏️ **Exercise: Match each outcome type to its canonical link: (a) a continuous, unbounded outcome, (b) a count, (c) a binary outcome. Choose from: identity, log, logit.**

# Concept: Iteratively Reweighted Least Squares (IRLS)

- GLMs have no closed-form solution (except OLS) — fitting is iterative
- Each step solves a weighted least squares problem on a "working response" z
- Weights come from the variance function; they change every iteration
- IRLS is Newton-Raphson / Fisher scoring applied to the GLM likelihood

✏️ **Exercise: Why can't Poisson regression be solved in one step the way OLS can, even though both use a linear predictor $X\beta$?**

# Concept: Deviance

- Deviance generalizes the residual sum of squares to any GLM family
- D = 2 × (log-likelihood of the saturated model − log-likelihood of the fitted model)
- Saturated model: one parameter per observation, a perfect (but useless) fit
- Smaller deviance = better fit, analogous to smaller SSE in OLS

✏️ **Exercise: Why is the saturated model — which fits the data perfectly — not itself a useful model, even though it appears in the deviance formula as the benchmark?**

# Concept: Poisson Regression for Counts

- Models counts (0, 1, 2, ...) with mean $\mu$ using the log link: $\log(\mu) = X\beta$
- Poisson assumption: variance equals the mean, $Var(Y) = \mu$
- $\exp(\beta_j)$ is the multiplicative change in the expected COUNT per unit of $x_j$

✏️ **Exercise: A Poisson regression coefficient for "years of experience" is $\beta = 0.05$. Compute $\exp(\beta)$ and interpret it as a percentage change in expected count per additional year.**

# Concept: Offsets and Exposure

- An offset converts a count model into a RATE model: $\log(\mu) = \log(\text{exposure}) + X\beta$
- The offset's coefficient is fixed at exactly 1 — not estimated
- Without an offset, a model can't distinguish "more events" from "more time observed"

✏️ **Exercise: Hospital A sees 40 infections in 1000 patient-days; Hospital B sees 30 infections in 500 patient-days. Without using an offset for patient-days, would a raw Poisson count model correctly identify which hospital has the worse infection RATE?**

# Concept: Overdispersion and the Negative Binomial

- Overdispersion: observed variance exceeds what the Poisson assumption (Var=mean) predicts
- Common causes: unmeasured heterogeneity, clustering, excess zeros
- Negative binomial adds a dispersion parameter, allowing $\mathrm{Var}(Y) = \mu + \alpha\mu^2$
- Ignoring overdispersion understates standard errors and overstates significance

✏️ **Exercise: A Poisson model's residual deviance is 350 on 200 degrees of freedom. Is this consistent with the Poisson assumption? What would you consider doing instead?**

# Concept: Binomial Regression for Proportions

- Models grouped proportions: k successes out of n trials per row
- Different from logistic regression on individual 0/1 outcomes only in data shape
- Uses the same logit link and Bernoulli-family likelihood, weighted by n

✏️ **Exercise: A dataset records, for each of 50 factories, the number of defective units out of that factory's daily production run. Why is grouped binomial regression more appropriate here than treating each unit as its own row in a plain logistic regression?**

# Concept: Gamma Regression for Positive Continuous Outcomes

- Models strictly positive, continuous, often right-skewed outcomes (cost, duration)
- Typically paired with a log link: $\log(\mu) = X\beta$
- Gamma's variance is proportional to $\mu^2$ — larger predicted means have larger absolute spread

✏️ **Exercise: Insurance claim amounts are always positive, right-skewed, and larger claims tend to have much larger variability than small claims. Why does this description point toward Gamma regression rather than OLS on the raw claim amount?**

# Concept: Model Diagnostics — Pearson and Deviance Residuals

- Raw residuals ($y - \hat{\mu}$) aren't comparable across GLM families with different variance structures
- Pearson residual: ($y - \hat{\mu}$) / sqrt($V(\hat{\mu})$) — standardized by the variance function
- Deviance residual: signed square root of each observation's deviance contribution
- Both should show no pattern against fitted values if the model is well-specified

✏️ **Exercise: Why can't you just plot raw residuals ($y - \hat{y}$) for a Poisson model the same way you would for OLS, and expect them to look randomly scattered even when the model fits well?**

# Concept: Model Comparison — AIC, BIC, and Likelihood Ratio Tests

- AIC = -2×log-likelihood + 2×(number of parameters) — lower is better
- BIC penalizes extra parameters more heavily for large samples
- Likelihood ratio test compares two NESTED models via their deviance difference
- Only compare AIC/BIC across models fit to the exact same data and same response

✏️ **Exercise: Model A (3 parameters) has log-likelihood -200. Model B (5 parameters, nesting Model A) has log-likelihood -197. Compute AIC for both. Does the extra complexity in Model B look worthwhile by AIC?**

# Concept: Regularized GLMs

- Ridge/lasso/elastic-net penalties extend directly to the GLM log-likelihood
- Same motivation as Chapter 1: control overfitting, handle many/correlated predictors
- Penalty is added to the negative log-likelihood, not the sum of squared errors
- Feature standardization is still required before penalizing

✏️ **Exercise: You are fitting a logistic regression with 500 candidate predictors and only 300 observations. Why is ordinary (unpenalized) maximum likelihood estimation likely to fail here, and what class of fix does this chapter recommend?**

# Concept: Separation in Logistic Regression

- Complete separation: a predictor (or combination) perfectly predicts the outcome
- The MLE for that coefficient runs off to infinity — the algorithm may not converge
- Common with rare events, small samples, or too many predictors
- Fix: Firth's penalized-likelihood correction, or a regularized/Bayesian approach

✏️ **Exercise: A logistic regression coefficient's estimate is 45.2 with a standard error of $8.1\times10^6$. What does this combination strongly suggest is happening, and why would simply reporting this coefficient's p-value be misleading?**

# Concept: GEE vs. GLMM for Clustered Data

- Ordinary GLMs assume independent observations — often false for repeated/clustered data
- GEE (Generalized Estimating Equations): estimates population-average effects
- GLMM (Generalized Linear Mixed Model): estimates cluster-specific effects via random effects
- The two give numerically different coefficients for non-linear links — both are "correct", answering different questions

✏️ **Exercise: A researcher wants to know "on average across the population, how does the treatment change the probability of recovery?" versus a doctor who wants to know "for THIS specific patient, given their cluster's random effect, how does treatment change their probability?" Which approach (GEE or GLMM) answers each question?**

# Concept: Calibration vs. Discrimination

- Discrimination (e.g. ROC-AUC): does the model rank higher-risk cases above lower-risk ones?
- Calibration: does a predicted probability of 0.3 actually correspond to ~30% observed frequency?
- A model can discriminate perfectly while being badly calibrated, and vice versa
- Calibration plots (predicted vs. observed by bin) reveal problems ROC-AUC cannot

✏️ **Exercise: A model achieves an excellent ROC-AUC of 0.90, but every predicted probability is roughly double the true observed rate in each risk bin. Has this model failed at discrimination, calibration, or both? Is it safe to use its raw probabilities for a business decision that depends on the actual probability value (not just the ranking)?**

# Mastery Check: Are You Ready to Move On?

*If you can answer every question below confidently, you have mastered Chapter 2.*

1. Name the three components of a GLM and explain what each one contributes.
2. Explain why OLS is a special case of the GLM framework rather than a separate method.
3. Explain why GLM fitting requires an iterative algorithm (IRLS) instead of a closed-form solution.
4. Define deviance and explain what the saturated model represents in that definition.
5. Explain the role of an offset in a Poisson rate model, and why its coefficient is fixed at 1.
6. Explain what overdispersion is, how to detect it, and two ways to correct for it.
7. Explain why Pearson or deviance residuals are used instead of raw residuals for GLM diagnostics.
8. State the formula for AIC and explain what it trades off against BIC.
9. Describe the signature (in coefficients and standard errors) of separation in logistic regression, and one fix.
10. Explain the difference between a GEE population-average effect and a GLMM cluster-specific effect.
11. Explain the difference between discrimination and calibration, with an example where they diverge.
12. Fit a Poisson regression with an offset in statsmodels and interpret exp(beta) correctly.
13. Explain why regularizing a GLM's log-likelihood requires the same feature standardization as Chapter 1's ridge/lasso.
14. Give an example of an outcome type well-suited to Gamma regression, and explain why OLS would be a poor fit for it.